between \$540 and \$707 to enter this lottery. These estimates come from a very large body of experimental and survey evidence. Some of the papers in the literature are quite creative. Andrew Metrick (1995) studied participants on the game show Jeopardy. The stakes on this show are high, and there is tremendous pressure because only the winning player gets to return. (Jeopardy participants are also really smart, so you can't say that they don't know what they're doing!) Metrick finds that risk aversion levels are low and near risk-neutrality. More recent estimates are provided by Aarbu and Schroyen (2009), who surveyed Norwegians and found risk aversion estimates around four. Kimball, Sahm, and Shapiro (2008) also perform surveys and their estimates are higher, at around eight. Note that these are surveys, however, rather than real financial choices. At Columbia Business School, Paravisini, Rappoport, and Ravini (2010) examined actual financial decisions made by investors in an online person-to-person lending platform deciding who gets funding, and how much. They estimated investors have risk aversions around three.

2.4. EXPECTED UTILITY

Expected utility combines probabilities of outcomes (how often do these events occur?) with how investors feel about these outcomes (are they a bad time for me?):

$$U = E[U(W)] = \sum_s p_s U(W_s). \tag{2.2}$$

In equation (2.2), the subscript s denotes outcome s happening, so expected utility multiplies the probability of it happening, p_s, with the utility of the event in that state, $U(W_s)$. Technically, we say that expected utility involves *separable* transformations on probabilities and outcomes. The probabilities measure how often bad times occur, and the transformations of outcomes capture how an investor feels about these bad times.

Expected utility traces back to the 1700s to the mathematicians Gabriel Cramer and Daniel Bernoulli, but its use as a decision-making tool exploded after John von Neumann and Oskar Morgenstern formalized expected utility in *The Theory of Games and Economic Behavior* in 1944. Von Neumann was one of the greatest mathematicians of modern times, and he also made seminal contributions in physics and computer science. He and the economist Morgenstern created a new field of economics—game theory—which studied how agents interact strategically, as in the famous Prisoner's Dilemma game. Von Neumann and Morgenstern initially developed expected utility to apply to games of chance; expected utility theory was just an accompaniment to the overall theory of games. The stock market is a game of chance—but unlike poker, stock market probabilities are ultimately unknown. In an important extension, Savage (1954) showed that expected